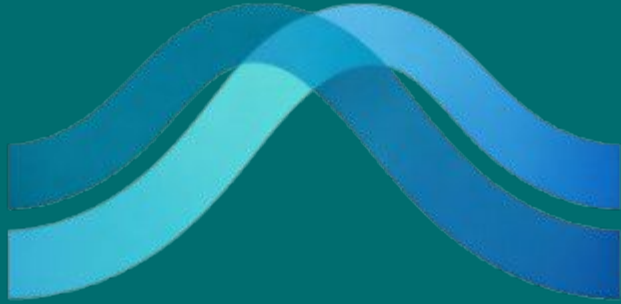




GRADUATION PROJECT • EXECUTIVE PITCH

The Bridge

Realtime Two-Way Sign Language Interpreter

Presented to **Prof. Dr. Gihan Almenyawy** · **Dr. Alsaid Essam** & to Whom it may Concern

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Locked out of their own society.

“ **1.2 million Egyptians** rely entirely on Egyptian Sign Language — a language completely distinct from spoken Arabic — to navigate their daily lives.

They are functionally locked out of the labor force, higher education, and foundational healthcare.



Medical misdiagnosis risk



Education inaccessibility



Excluded from the
workforce

A crisis hiding in plain sight.

16.02%

of Egypt's population lives with
hearing loss

1.2M

are profoundly deaf and depend on
Egyptian Sign Language

≈ 0

scalable automated solutions for
EGSL exist today



The core bottleneck: a severe, systemic shortage of specialized human sign-language interpreters — institutions simply cannot solve this by hiring more people.

Communication gaps cost lives, livelihoods, and learning.



Healthcare

Written notes fail due to high illiteracy rates; relying on family translators leads to breaches of confidentiality and severe misdiagnosis risk.



Higher Education

Deaf students are functionally excluded from lectures, exams, and academic participation — despite clear legal protections.



Employment

Companies avoid hiring deaf talent because day-to-day workplace communication is unmanageable at scale.



Financial Services

Deaf customers cannot independently transact with banks or government services — a matter of basic dignity and autonomy.

Accessibility is no longer optional — it is the law.



Law No. 10 of 2018

Higher Education Mandate

Universities are legally prohibited from rejecting students on the grounds of disability and must provide "reasonable facilitating means."

Officials face direct criminal liability for non-compliance.



5% Employment Quota

Corporate Accessibility Mandate

Any firm with 20 or more workers is mandated to maintain a 5% employment quota for persons with disabilities.

Non-compliance carries severe legal and financial penalties.

Institutions need a scalable solution — **The Bridge turns legal obligation into a cost-effective opportunity.**

Why every existing option fails the Egyptian deaf community.



Wrong language

Western Apps

(ASL / BSL platforms)

Global platforms are trained on American or British Sign Language — functionally useless for 1.2M EGSL users in Egypt.



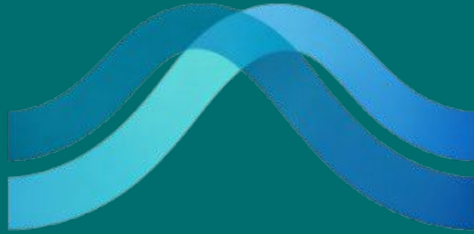
Doesn't scale

Local Human-in-the-Loop

(Wasel, Merge)

Essentially call centers relying on scarce, expensive human interpreters. Cannot handle continuous daily use — lectures, triage, meetings.

INTRODUCING



The Bridge

A fully-automated, real-time communication platform — purpose-built for Egyptian Sign Language and spoken Arabic.



Two-way

EGSL ⇌ Arabic



Instant

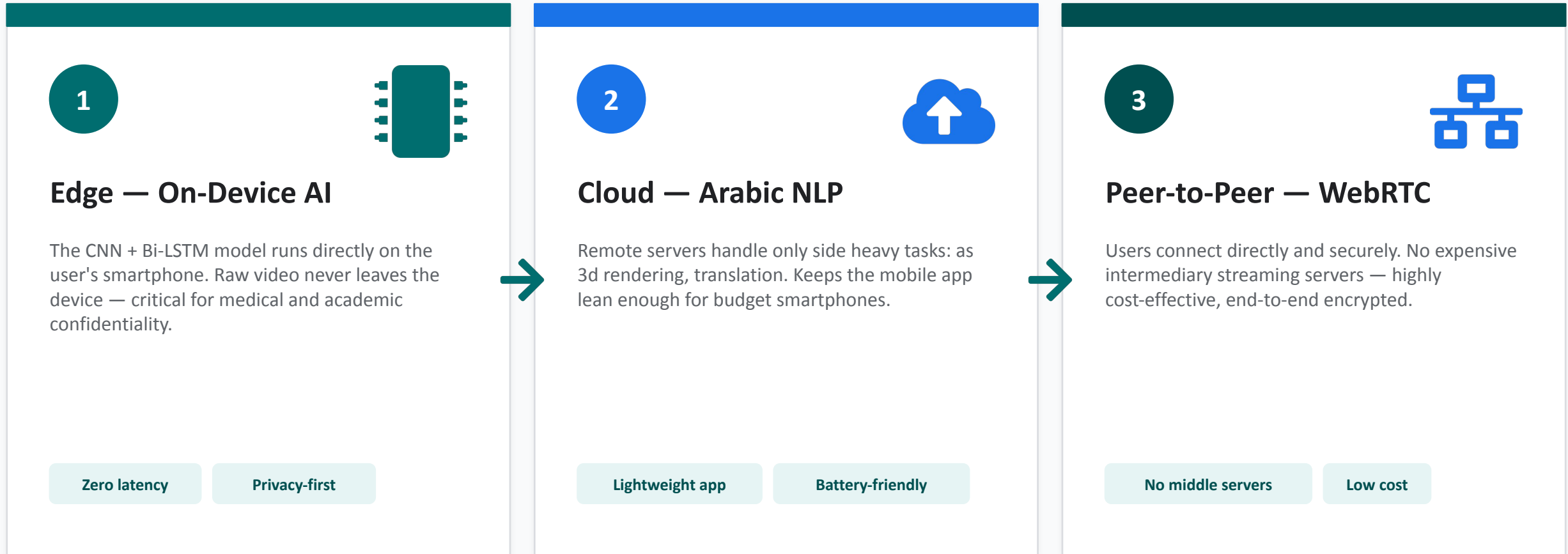
On-device, zero delay



Scalable

No human in the loop

A hybrid design that is fast, private, and cheap to run.

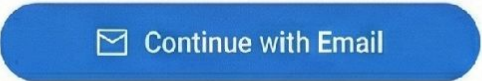


Screen A: Onboarding & Login



The Bridge





User Preferences

Deaf

Hearing

Preferred Sign Language

ASL

EGSL

Select your preferred sign language for better matching.

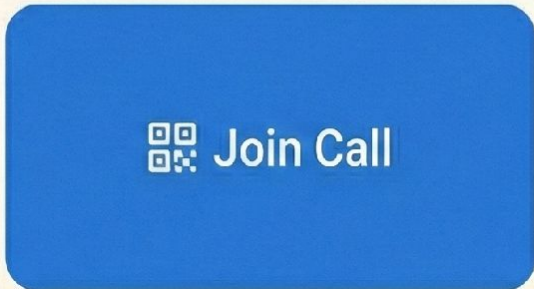
Accessibility Options

Screen B: Home Dashboard

Home









Home

History

Profile

Screen C: Active Video Call



[Interpreter]: The appointment is confirmed for next Tuesday at 10:00 AM. Is there anything else?

Mute

Camera Flip

Text Mode

End Call

The System Features

**Real-Time Video
Call Interpreter**

**Interpret Visual Signs
into Text/Transcription**

**Voice to Sign
Language Encoding**

Speech-to-Text Ability

**Support for Multiple
Languages (US & EG Sign
Language)**

**Translation Between Sign
Languages (US ↔ EG)**

Future Scope: The future scope includes the 3d rendering, and support for other Sign Languages

Our stack — chosen for accuracy, speed, and reach.



Flutter

Cross-platform mobile app



MLP

Static sign recognition



Bi-LSTM

Dynamic gesture sequences



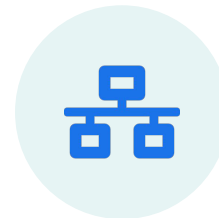
Python

Model training & inference



NestJS

Backend orchestration



WebRTC

P2P live video calls

Our core innovation: a custom hybrid CNN + Bi-LSTM model that fuses static sign shapes with dynamic gesture sequences for natural EGSL translation.

One engine. Four sectors. Nationwide reach.



Higher Education

Universities deploy The Bridge as the default accessibility layer across every faculty — meeting Law 10/2018 without the prohibitive cost of human interpreter contracts.

Pilot target: Cairo University faculties



Healthcare & Telehealth

Hospitals and telehealth platforms integrate our SDK for patient triage and consultations — mitigating malpractice risk and protecting patient privacy.

Integration target: Vezeeta, public hospitals



Workplace & HR

Companies integrate deaf hires seamlessly, unlocking the 5% quota without ongoing interpreter salary costs — plus a 50% tax exemption uplift from government.

Target: Firms with 20+ employees



Financial Services

Banks deploy The Bridge on branch tablets to serve deaf customers — satisfying recent Central Bank of Egypt accessibility directives.

Target: Retail banks, ATMs

Free for the deaf! Revenue from those legally required to serve them.

Subsidized freemium core: Free for individual deaf users to drive adoption — which simultaneously generates the training data that improves the model.



STREAM 1

B2B Enterprise SaaS

Annual licenses to universities, banks, and hospitals — dramatically cheaper than hiring human interpreters to meet legal mandates.



STREAM 2

CSR Partnerships

Egyptian banks alone allocated EGP 136M to disability CSR recently. Major institutions underwrite server costs in exchange for in-app visibility.



STREAM 3

API / SDK Integration

Telehealth platforms, e-commerce, and customer service portals embed our engine for recurring licensing fees.



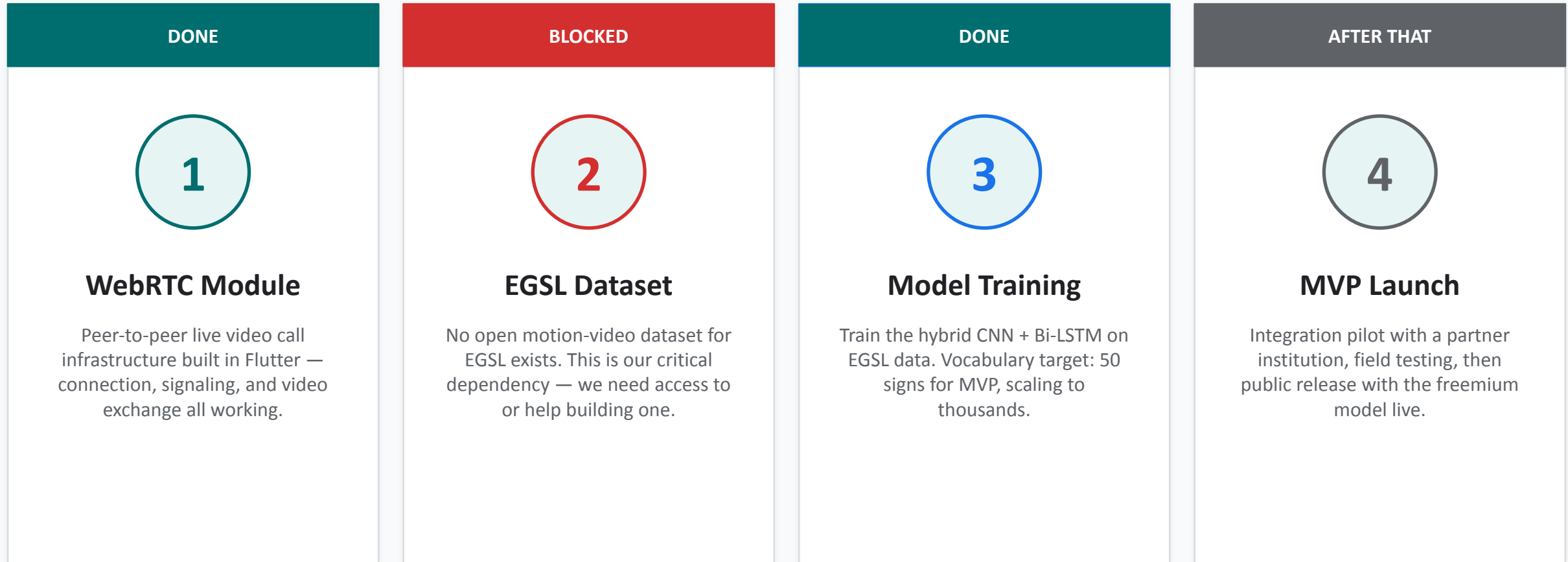
STREAM 4

Premium B2C

Students and professionals upgrade for specialized vocabularies (medical, engineering) and offline translation. Low price, high LTV.

Bonus incentive: Government offers a **50% tax exemption uplift** for companies exceeding disability employment quotas — a powerful pull factor for corporate adopters.

Where we are, and what we need to ship.



 **Timeline pressure:** Without dataset access, the graduation deadline forces us into a much smaller scope than the idea deserves.

What we need from you to bring The Bridge to life.

ASK #1 — DATA ACCESS



Dynamic EGSL Dataset

Only static EGSL data exists publicly. Motion-video data — equivalent to ASL's WLASL — is the missing foundation for real-world EGSL translation.

PLAN A

Access the Ministry of Education's existing 12,000-word dynamic EGSL dataset — ideally open-sourced for the wider research community.

PLAN B (if Plan A is not feasible)

Facilitate a collaboration with the Deaf Institution at Cairo University to co-build a starter 50-word EGSL motion dataset — enough for a working MVP.

ASK #2 — SUSTAINABILITY



Funding & Partnerships

Help us secure long-term runway by connecting us to institutional funders and private-sector sponsors who can underwrite the project beyond graduation.

- ✓ University-facilitated sponsorship channels
- ✓ Introductions to corporate CSR programs
- ✓ Support for future grant applications

With your support, The Bridge moves from a graduation project to a national infrastructure for the deaf community.

Contact Us



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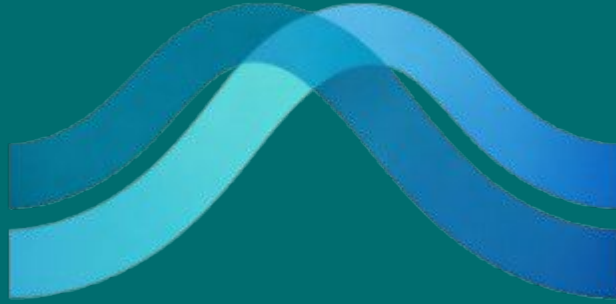
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Thank you.

We're excited to bridge this gap — with you.